Manual

PCC Module: Scale Interfacing

PCC-WAK 8.1

Version 1.0.23049

Last changed on: 02.09.2020

Copyright

Contents

1 Overview of the PCC Module: Scale Interfacing

Purpose

Interface modules in order to communicate with scales (e.g. transferring weighing data)

Implementation notes

You use the function package if:

- you want to connect a scale to the MES using interfaces.
- The modules required to implement the protocol must exist or be established specifically to the machine. Specifications are defined during machine installation counseling.

Integration

The protocol module has been integrated in HYDRA-PCC. Please also see the documents dealing with the products SCS-PCP and SCS-PCB.

Functions

Transfer of weighing data of different scale manufacturers

2 PCC Module Connecting Scales

Scale Driver Bizerba ST

2.1.1 General

Description of the scale driver Bizerba ST. The connection is established via a special RS232 cable. Scale values are requested using special terminal dialogs.

The described PCC driver supports scales of the type Bizerba ST.

Required files:

- •bizerba_st.dll
- •bizerba_st.ini

2.1.2 Interfaces

Using the RS232 connection, the host computer is directly connected with the scale via a serial cable.

Contact:	Short description	Meaning
2	RxD	Receive data
3	TxD	Transmit data
5	GND	Signal ground

Default interface settings are:

- 9600 Baud
- 7 data bits
- 1 start bit
- E parity bits
- 1 stop bit

There is **no** handshake. It must be guaranteed that the connected host computer is always able to hold a complete telegram in its receive buffer.

2.1.3 Description of telegram Bizerba ST

New description of		(including calibratable/verifiable	

telegram: Bizerba scale		memory)	
Start position	**Name/Description**	**Length**	**Example/Explanation**
1	ID	1	**]**
2	Scale number	1	**1** (scale 1, fixed)
3	ID	2	**Z0** (please note: additional field)
5	Reference number	6	**000123** (right-aligned, filled with 0 (zero))
11	ID	1	**+** (plus sign. Status should not be evaluated)
12	Status	1	Status (should not be evaluated)
13	Gross weight	11	**xxxxx23.4kg** (right-aligned, filled up with blank characters)
24	ID	1	**.** (normally dot or / (forward slash)
25	Tare weight	10	**xxxx23.4kg** (right-aligned, filled up with blank characters)
35	ID	1	**,** (comma)
36	Status	1	Status (should not be evaluated)
37	Sign	1	Blank character with positive weight, minus (-) sign with negative weight

38	Net weight	10	**xxxx99.5kg** (right-aligned, filled up with blank characters)

LENGTH OF TELEGRAM: 47

Sample string:

char(#2) +']1Z0001209+!   230.0kg   005.0kg+!   225.0kg' + char(#3)

Gross = 230 kg

Tare = 5 kg

Net = 225 kg

2.1.4 Settings bizerba_st.ini

[WAAGE_001] ➔ ID for the driver section

COM=1,9600,7,E,1 ➔ Settings for the COM port
e.g. COM=1,9600,7,E,1
COM port = 1
Baud = 9600
Data bit = 7
Parity = E
Stop bit = 1

V:EGR:GUTS=BRUTTO ➔ ID for the scale values of the dialogs.
Possible settings:
BRUTTO (GROSS)
NETTO (NET)
TARA (TARE)
EINHEIT (UNIT)

The following IDs are sent as of version 7.2.1.8 /25 April 2014

These parameters are set along with the new processing

SCALE_MODE=1

- Reference number of the scale
- ID number of the scale

The following IDs are always sent automatically:

V:WAAGE:REFNR ← Reference number

V:WAAGE:NR ← ID number

Data record sent by the scale:

]1{53291{5654321+) 003,8kg 000,0kg+) 003,8kg'#3

Example:

ID number of scale	= 291
Reference number of scale	= 654321

2.1.5 Configuration options for data collection

The positions and lengths for reading out data from the scale string can still be changed.

Notes on the configuration structure

Value that is read out = position within the string, number of characters to be read

IDs for which the position and number of characters can be changed:

1st value corresponds to the string position; 2nd value represents the length of characters to be read out

- POS_BRUTTO
- POS_NETTO
- POS_TARA
- POS_EINH
- POS_REFNR
- POS_IDENT

By default, the following positions are defined:

- POS_BRUTTO=19,9
- POS_NETTO=43,9

- POS_TARA=31,9
- POS_EINH=28,2
- POS_REFNR=11,6
- POS_IDENT=8,3

Additional configurations:

REFERENZ_NR=OFF	(no reference number is sent in order to be collected)
IDENT_NR=OFF	(no scale ID number is sent in order to be collected)

2.2 Scale Driver Busch Minipond

2.2.1 General

The scale unit Busch Minipond consists of an indicator and evaluation module to connect different weighing cells. The indicator unit including display and function keys allows for the scale unit to be configured and operated.

There is a serial RS-232 interface to communicate with the HYDRA system. It is possible request weight values and to transfer setting parameters to the scale controller.

Required files:

- minipond_25.dll
- minipond_25.ini

2.2.2 Example for driver INI

Example:

```
[SERVICE]
info=minipond_25.dll
tracing=0
TraceLevel=5
LogLevel=2
TIMEOUT=180

[WAAGE_000]
COM=1,19200,7,E,1
```

RTS-CONTROL=ON

RESET_SCALE_AT_BEGIN=OFF

SET_DISPLAY_ZERO_AFTER_READ=ON

V:WAAGE:BRUTTO=BRUTTO

V:WAAGE:NETTO=NETTO

;V:WAAGE:NETTO=DISPLAY

;V:WAAGE:DISPLAY=DISPLAY

V:WAAGE:TARA=TARA

POLL=0

POLL_I=500

2.2.3 Settings: minipond_25.ini

The serial transmission properties need to be defined in order to communicate with the Busch scale controller. Speed and control characters are defined via an INI entry:

COM=1,19200,7,E,1

COM=<Nr>,<Bd>,<Dat>,<Parity>,<Stop>

<Nr> Number of the serial interface

<Bd> Baud rate

<Dat> Number of data bits

<Parity> Parity (Even, Odd, None, Mark)

<Stop> Number of stop bits

In the above example HYDRA communicates via the interface COM1 at a transmission rate of 19200 Bd and a character length of 7 bits. Parity is set to "even" and the data word is completed with 1 stop bit.

Subject to the laying of the serial connection cable, handshake processing must be activated via the RTS signal. The following entry is required:

RTS-CONTROL=ON

The following diagram shows how an industrial PC can be connected to the scale using serial communication (RS-232).

RS232 PC connection cable with 9-pole D-Sub socket (article No. 10KAB202 / ST.2300.0019)

Terminal		Color		PC
TxD	5	Green	2	RxD
RTS	6	Yellow	8	CTS
RxD	7	Brown	3	TxD
CTS	8	White	7	RTS
GnD	1	Gray	5	GnD
			1	
			4	
			6	

Request weight

To request the weight, it can be differentiated between:

- Gross weight V:WAAGE:BRUTTO=BRUTTO
- Net weight V:WAAGE:NETTO=NETTO
- Tare weight V:WAAGE:TARA=TARA
- Display V:WAAGE:NETTO=DISPLAY

The parameter "Display" can be used to take over the currently displayed value. This value depends on taring of the scale. If, apart from taking over the display value, the parameter

SET_DISPLAY_ZERO_AFTER_READ=ON

is also set, taring restarts automatically after taking over the value and the displayed value is reset to "0.0". The next time an item is weighed, the value is transferred to HYDRA and added up there.

The scale only transfers weight values if they are stable. This means, unstable loads first have to become steady.

Please retry if no value is taken over after requesting the weight.

Communication disturbances

TIMEOUT=180

The value configured in "TIMEOUT" defines a period of time for the scale to send a response. The value 180 corresponds to a period of approx. 10 seconds. The scale driver sends an error message (transmission error) including the following contents to HYDRA if the response to requesting the weight is not available within this period of time:

Example: the scale driver's response when requesting the net weight and communication fails:

DLG=EVENT|DRV= minipond_25.dll|DRVINST=WAAGE_000|V:NETTO=TIMEOUT|

Initialization of the display unit

The scale can be re-initialized along with starting the terminal application in CT-WIN. If the parameter

RESET_SCALE_AT_BEGIN=ON

is used, CT-WIN sends once the initialization commands to the scale controller.

This restores the initial state defined by the scale manufacturer.

2.2.4 Log files

The driver program of the scale connection records communication with the scale in a corresponding log file within the "spool" sub-directory of the terminal application.

The item "LogLevel=" configures the number of recorded events. If this value is increased (max. 5), the number of recorded messages will also increase.

The file is prevented from increasing excessively by self-regulating checking and overwriting of data.

2.2.5 Attachment

The attachment includes the communication profiles of the Minipond 25 scale unit.

Initialization of the interface and communication

Meaning	**Command**	**Return value**	**Meaning**
Initialize Device	STX i D ETX		
Initialize Communication	STX i C ETX	STX<Err>ETX	Err=EC_OK (communication enabled) Err=EC_COM (communication disabled/blocked)
Synchronization	STX ETX	STX<Err>ETX	Err=EC_OK (slave has been synchronized)

Command requesting weight values

Meaning	**Command**	**Return value**	**Meaning**
Zeros	STX z ETX	STX<Err>ETX	Err=EC_OK (zeroing/resetting to zero successful) Err=EC_COM (zeroing/resetting to zero failed)
Tare	STX t ETX	STX<Err>ETX	Err=EC_OK (taring successful) Err=EC_COM (taring failed)
Reset to gross	STX g ETX	STX<Err>ETX	Err=EC_OK (untaring

			successful) Err=EC_PAR (untaring failed)
Read weight value (displayed value)	STX L D ETX	STX<Err><E_Code><Stat><WB><Gew>ETX <E_Code>: error of weighing system (see section 4 of manual) <Stat>: weight status (see section 4 of manual, weight status) <WB>: weighing range (1-3) <Gew>: weight/kg 8 characters (example:__120,60)	Err=EC_OK (weight information available) Err=EC_PAR (weight not available)
Read gross value (displayed value)	STX L G ETX	STX<Err><E_Code><Stat><WB><Gew>ETX <E_Code>: error of weighing system (see section 4 of manual) <Stat>: weight status (see section 4 of manual, weight status) <WB>: weighing range (1-3) <Gew>: gross value/kg 8 characters (example:__120,60)	Err=EC_OK (gross value available) Err=EC_PAR (gross value not available)
Read net value (displayed value)	STX L N ETX	STX<Err><E_Code><Stat><WB><Gew>ETX <E_Code>: error of weighing system (see section 4 of manual) <Stat>: weight status (see section 4 of manual, weight status) <WB>: weighing range (1-3) <Gew>: net value/kg 8 characters (example:__120,60)	Err=EC_OK (net value available) Err=EC_PAR (net value not available)
Read tare value	STX L T ETX	STX<Err><E_Code><Stat><WB><Gew>ETX <E_Code>: error of weighing system (see section 4 of manual) <Stat>: weight status (see section 4 of manual,	Err=EC_OK (tare value available) Err=EC_PAR (tare value not available)

		weight status) <WB>: weighing range (1-3) <Gew>: tare value/kg 8 characters (example:__120,60)	
Read scale parameters	STX W r ETX	STX<Err><Scaling>ETX Scaling: scale parameters (see comment)	Err=EC_OK (valid scale parameters) Err=EC_PAR (invalid scale parameters)

Information included in the E_CODE of the scale reply

E_Code	**Description**
‘0‘	*No error in weighing system*
‘9‘	*Measured value out of range*
‘1‘	*Reference value out of range*
‘2‘	*Zero value out of range*
‘6‘	*Check number out of range*

The status byte <STAT>

Status byte	**Value = 0**	**Value = 1**
Bit 0	Weight valid	Weight invalid
Bit 1	Gross	Net
Bit 2	Unit kg	Unit Lb
Bit 3	Weight within weighing range	Weight out of weighing range
Bit 4	No stability	Stability

Bit 5	Weight above minimum load	Weight below minimum load
Bit 6	Always "1" --> status is printable character	
Bit 7	Always "0“ → matches 7 Bit ASCII encoding	

Error code <Err> included in the scale's response message

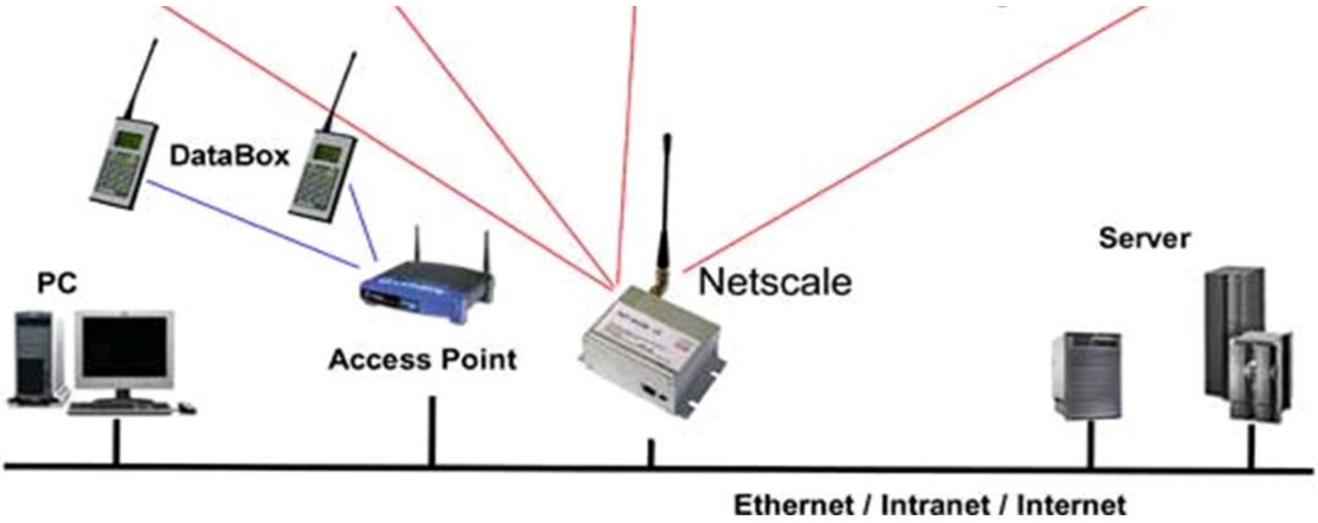
Code <err>	Name/Description	Meaning	Note
‘0‘	EC_OK	Command executed correctly	
‘1‘	EC_IC	Command interface not initialized	This return value enables the master to detect any reset and to trigger initialization. After restarting the slave, the master has to open the interface using the command "Initialize Communication".
‘2‘	EC_NY	Command processing not yet completed	Sent as reply for the DIN measurement bus protocol if the requested data are not yet available. Does not occur with point-to-point connection.
‘3‘	EC_COM	Command cannot be executed.	Command detected but cannot yet be executed.
‘4‘	EC_PAR	Invalid or wrong command parameters	
‘7‘	EC_FKT	Unknown command	

2.4 Scale driver crane scale EHP UDP protocol NSG G.1

2.4.1 General

The crane scale communicates via radio with a net scale central unit. The data of individual crane scales can be read out by UDP protocol. The shop floor terminal selects the corresponding crane scale by configuration.

Scale 1 Scale 2 Scale 3 **network-compatible** up to 20 devices



Required files:

- EHP_Netscale_UDP.dll
- EHP_Netscale_UDP.ini

2.4.2 Example for driver INI

[SERVICE]

info=EHP_Netscale_UDP.dll

intervall=0

testmode=0

tracing=0

TIMEOUT=180

TraceLevel=5

ExecuteQueue=1

ThreadBaseId=100

ShowErrorWindow=OFF

[WAAGE_001]

Waage_IP=192.168.100.71

Waage_Port=187

Waage_ID=1

MaxRepeatIfError=20

TryAgainTimeAfterError=500

Weight_Read_Modus=DOUBLE

V:Waage1:BRUTTO=BRUTTO

2.4.3 Settings of EHP_Netscale_UDP.ini

UDP interface

The network address and port have to be defined in order to establish communication with the Netscale scale controller. They are defined via INI entries:

Waage_IP=192.168.100.71

Waage_Port=187

Since several scales can be connected to this scale controller, this configuration applies for all scales within the instance [Waage_001]. Therefore, individual crane scales are differentiated as follows:

V:Waage1: …

V:Waage2: …

Request weight mode

The log file of the EHP scale controller Netscale differentiates between two different ways of requesting the weight.

Single weight request

The single request for weight is defined in the INI file by the entry:

Weight_Read_Modus=SINGLE

If this mode is selected, the scale only returns one weight value. It has to be defined in the configuration of scales if this value refers to the gross or net weight. HYDRA cannot differentiate this.
This means that the same result is delivered although the tare weight is defined if weight is requested using the entries: Waage1:BRUTTO=BRUTTO and V:Waage1:NETTO=NETTO

Extended weight request

If the extended request for weight is selected, the scale controller transmits two weight values. Consequently, it is possible to request the net, gross and/or tare weight from HYDRA. The following settings are possible:

V:Waage1:BRUTTO=BRUTTO

V:Waage1:NETTO=NETTO

V:Waage1:TARA=TARA

The INI file can be configured accordingly by choosing the request mode: DOUBLE

Weight_Read_Modus=DOUBLE

Communication disturbances

If a coil is transported using a crane scale, its weight is not stable as it sways. To avoid faulty measurements, no weight values are transferred if the weight is instable. Consequently, it might be the case that there will be no result when HYDRA asks for the weight and the request dialog needs to be restarted. Additionally, communication with the scale controller might be disturbed if the crane is moved outside of the reception range. Three IDs have been added facilitating the procedure:

MaxRepeatIfError=20

TryAgainTimeAfterError=500

ShowErrorWindow=OFF

MaxRepeatIfError=20 defines the number of retries and TryAgainTimeAfterError=500 specifies the time in ms after which a retry is triggered. If both values are multiplied, it results in the period of time the program attempts to get the weight (in this example: 20x500mS = 10 seconds).

If any attempt of getting the weight value within this period of time fails, it is possible to display a message window by using the ID ShowErrorWindow=ON.

This window remains open until it is closed by clicking OK. No message appears if ShowErrorWindow=OFF is set.

2.4.4 Log files

The driver program of the scale connection records communication with the scale in two different files in the local terminal directory.

The file "EHP_Netscale_UDP_WAAGE_00x.log“ records the load and unload activities, communication with the scale controller and with the higher-level HYDRA system.

Actual weight values sent and transferred by the scale controller can be displayed in the file "EHP_Netscale_UDP_WAAGE_00xWaage_Send.log".

The files are prevented from increasing excessively by self-regulating checking and overwriting of data.

2.5 Scale driver FILIZOLA ID-S

2.5.1 General

The scale unit FILIZOLA ID-S consists of an indicator and evaluation module to connect different weighing cells. All connections and interfaces are connected internally via cable connections thus allowing to be used even in the humidity range.

There is a serial RS-232 interface to communicate with the HYDRA system.

Required files:

- filizola_ids.dll
- filizola_ids.ini

2.5.2 Example for driver INI

```
[SERVICE]

info=filizola_ids.dll

tracing=0
```

TraceLevel=5

LogLevel=2

TIMEOUT=180

[WAAGE_000]

COM=1,9600,7,E,2

V:NETTO=NETTO

V:BRUTTO=BRUTTO

V:TARA=TARA

V:EINHEIT=EINHEIT

POLL=0

POLL_I=500

2.5.3 Settings of filizola_ids.ini

Serial interface

The serial transmission properties need to be defined in order to communicate with the FILIZOLA scale controller. Speed and control characters are defined via an INI entry:

COM=1,9600,7,E,2

In the above example HYDRA communicates via the interface COM1 at a transmission rate of 9600 Bd and a character length of 7 bits. Parity is set to "even" and the data word is completed with 2 stop bits.

Request weight

To request the weight, it can be differentiated between:

- Gross weight V:BRUTTO=BRUTTO
- Net weight V:NETTO=NETTO
- Tare weight V:TARA=TARA

The following entry provides the unit of the weight:

V:EINHEIT=EINHEIT

The scale should be configured in such a way that it only sends weight values on request by the connected system (transmission on demand).

For this purpose, the weighing system expects a command "ENQ" (05 Dec) and then transfers the data in the format:

"(STX)(SP)(SP)(X)(X)(X)(X)(X)(X)(X)(k)(g)(^)(T)(SP)(SP)(SP)(SP)(EXP)(N) (N) (N) (N) (N) (N) (N)(NEXP)(k)(g)(^)(L)(SP)(LF)(CR)(ETX)“

This means:

- (STX) Start of Text
- (SP) Space
- (X) tare (numbers + comma)
- (k) letter k of kg
- (g) letter g of kg
- (^) character circumflex
- (T) letter T
- (EXP) expansion character
- (N) net weight (numbers + comma)
- (NEXP) expansion cancel character
- (L) letter L
- (LF) "Line feed“
- (CR) “Carriage return”
- (ETX) End of Text

Communication disturbances

TIMEOUT=180

The value configured in "TIMEOUT" defines a period of time for the scale to send a response. The value 180 corresponds to a period of approx. 10 seconds. The scale driver sends an error message (transmission error) including the following contents to HYDRA if the response to requesting the weight is not available within this period of time:

Example: the scale driver's response when requesting the net weight and communication fails:

DLG=EVENT|DRV=filizola_ids.DLL|DRVINST=WAAGE_000|V:NETTO=TIMEOUT|

2.5.4 Log files

The driver program of the scale connection records communication with the scale in a corresponding log file within the "spool" sub-directory of the terminal application.

The item "LogLevel=" configures the number of recorded events. If this value is increased (max. 5), the number of recorded messages will also increase.

The file is prevented from increasing excessively by self-regulating checking and overwriting of data.

2.6 Scale Driver Mettler Toledo SICS

2.6.1 General

Description of the scale driver Mettler Toledo SICS. Connection is established via a special RS232 cable. Scale values are requested using special terminal dialogs (ctwin).

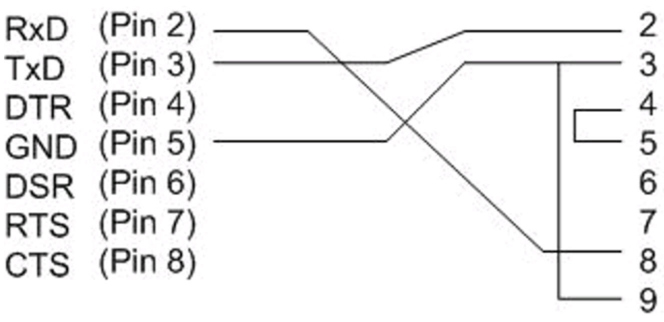
The described PCC driver supports scales by the manufacturer Mettler Toledo working with the SICS log file (e.g. 8142, Lynx, Panther).

2.6.2 Interfaces

Using the RS232 connection, the host computer is directly connected with the scale via a serial cable.

Contact:	Short description	Meaning
2	RxD	Receive data
3	TxD	Transmit data
5	GND	Signal ground

Shop floor PC scale E-1-TAD



Default interface settings are:

9600 Baud

7 data bits

1 start bit

E parity bits

1 stop bit

There is **no** handshake. It must be guaranteed that the connected host computer is always able to hold a complete telegram in its receive buffer.

2.6.3 Settings of mettler_sics.ini

[WAAGE_000]

COM=2,9600,7,E,1 ➔ Settings for the serial interface

// The scale's communication address

Scale-Type=Spider ➔ scale type:

- Lynx
- 8142
- Spider
-

// Request parameters specific to scale.

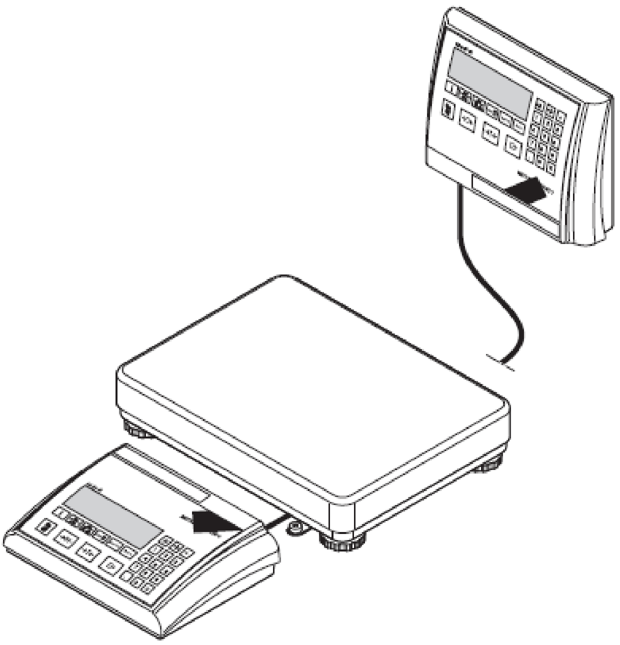
Brutto_Scale=SI ➔ gross value requested from scale

Netto_Scale=SI ➔ net value requested from scale

V:EGR:GUTS=BRUTTO ➔ Dialog ID including value requested from PCC. The following queries are possible:

- BRUTTO (GROSS)
- NETTO (NET)

2.6.4 Scale type Mettler Toledo SPIDER

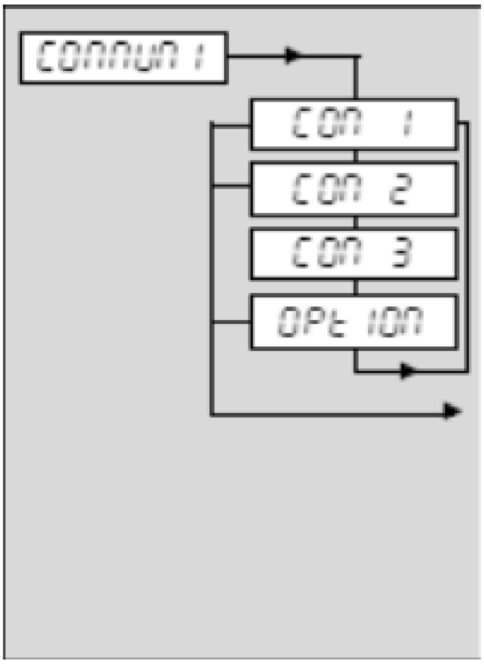


Scale configurations for server communication:

The operation mode of the interface must be set to **DIALOG.**

<u>Display</u> <u>Notes</u>

Configuration of scale interfaces: only accessible for supervisors!



Standard interface

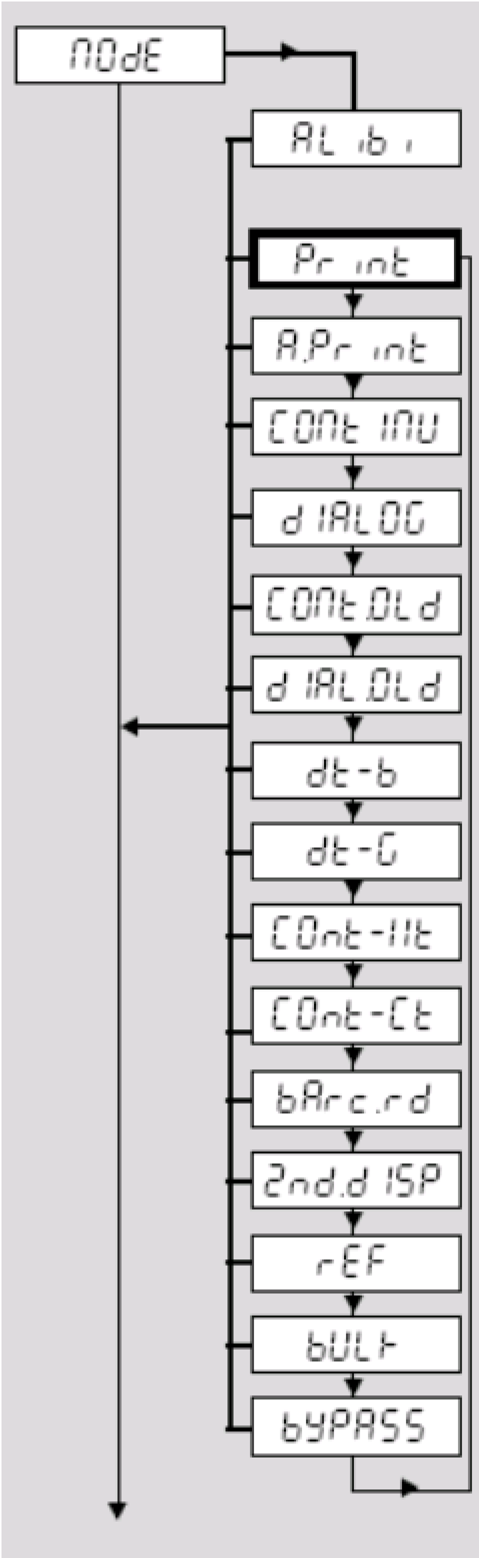
Optional interface

Optional interface

Analog option

Settings:

Operation mode of interface	→ section 4.7.1
Communication parameters	→ section 4.7.2
Settings for printing the protocol	→ section 4.7.3
Line feeds for protocol	→ section 4.7.4
Resetting interfaces	→ section 4.7.



Alibi memory, only appears if data transfer to the alibi memory is enabled (“APPLIC“→“Alibi.M“→“Transfer“→“ON“). Fixed setting (other operation modes only accessible if transfer function is disabled).

Manual data output at printer (key << >>. **Factory setting**

Automatic output of stable results at printer (for serial weighing mode)

Continuous output of all weight values via the interface. Not available for COM2, provided that analog option is enabled!

Bidirectional communication via MT-SICS commands (controlling the scale via PC) Not available for COM2, provided that analog option is enabled.

Like "continuous" (see above) but with 2 fixed blank characters in front of the unit (compatible with Spider 1/23)

Like "dialog" (see above). But scale sends 2 fixed blank characters in front of the unit (compatible with Spider 1/2/3)

Format compatible with DigiTOL Weight values to be transferred can be selected: tare, net, gross (gross weight is identified by "B").

Like“dt-b“ – mode (see above), But gross weight is identified by "G".

“TOLEDO Continuous Weight“ mode.

“TOLEDO Continuous Count“ mode.

Connecting a barcode reader.

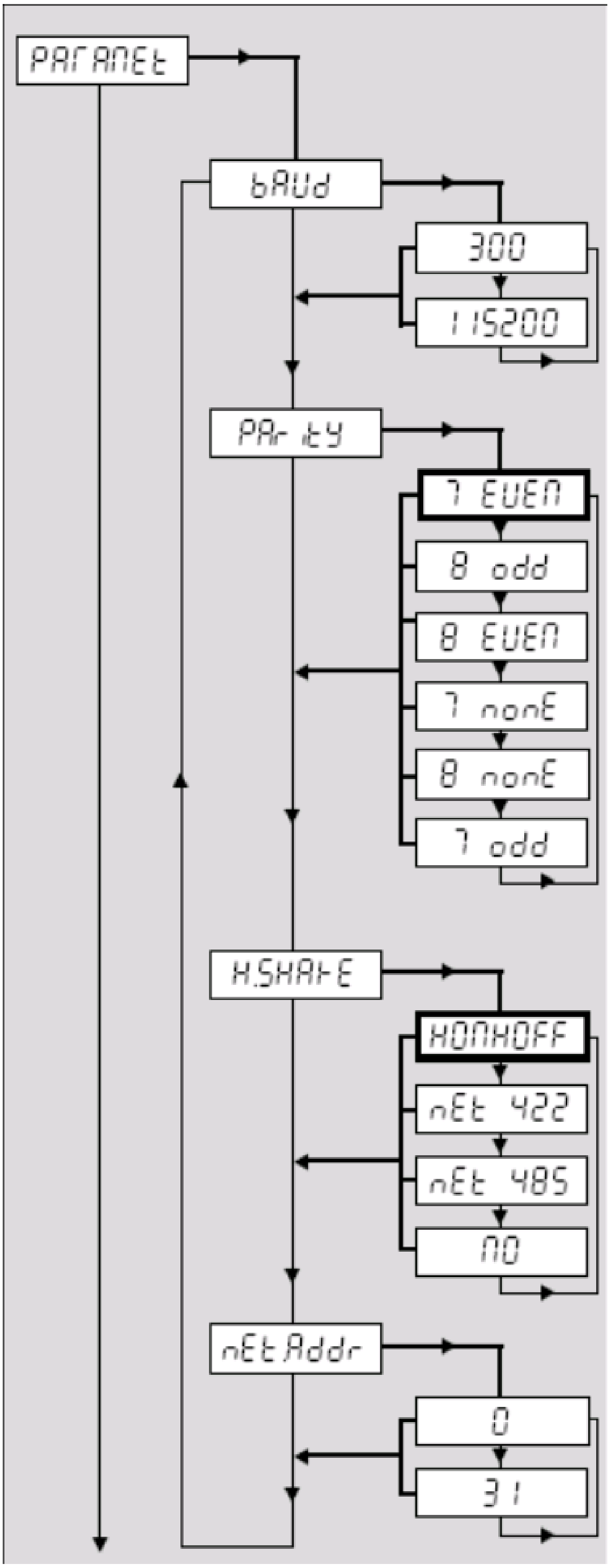
Connecting a second display. Not available for COM2, provided that analog option is enabled!

Second scale used as reference scale.

Second scale used as bulk scale.

Only for analog mode: disabling the analog option. If the "analog" option is not deactivated, the settings "Ref" and "Bulk" are not available for COM1 and COM3 (if available). The operation modes "Print" and "A.Print" are only available for COM2.

Display



Notes

Not available for analog option. The values set here must correspond with those of the connected peripheral devices (printers, PC).

Data transmission rate of the interface:

300 Baud – 115200 Baud. Factory settings depending on operation mode of the interface. **Please note:** The baud rates 57600 and 115200 are only available for COM3.

Number of **data bits** and **parity:**

7 data bits, even parity

8 data bits, odd parity

8 data bits, even parity

7 data bits, no parity

8 data bits, no parity

7 data bits, odd parity

Factory settings depending on operation mode of the interface.

Transmission log:

Xon/Xoff protocol (**factory setting**).

Network operation according to RS422 standard via optional RS422/485 interface (COM1). Not available for COM2/COM3.

Network operation according to RS485 standard via optional RS422/485 interface (COM1). Not available for COM2/COM3.

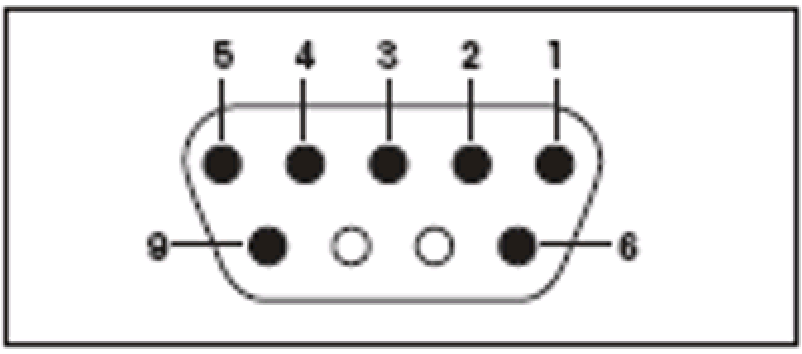
No communication protocol.

Network address (only available for “Net 422“ and “Net 485“. For further details on network operation see section 5.1.5).

Network addresses 0-31 are available.

Interface type:	Voltage interface according to EIA RS-232C/DIN 66020 (CCITT V24/V.28)
Max. cable length:	15m
Signal level data lines:	Level “0“ (with RL > 3 kΩ): +3V to +25V (high)
	Level “1“ (with RL > 3 kΩ): -3V to -25V (low)
Connections:	Spider: Sub-D, 9-pole, (female) connector, Spider S: round plug, 8-pole
Operation mode:	full duplex
Transmission type:	bit-serial, asynchronous
Transmission code:	ASCII
Protocol/flow control:	without, XON, XOFF
Baud rates:	300, 600, 1200, 2400, 4800, 9600
Data bits:	7 or 8
Stop bits:	Interface 1: receive at least 1 stop bit, send 2 stop bits
	Interface 2: always 1 stop bit for sending and receiving
Parity:	without, even, odd

Pin assignment RS232C Spider (view on connector)



Pin 1: digital output +5V/50mA, only interface 1

Pin 2: TxD (transmission line of the scale)

Pin 3:RxD (receiving line of the scale)

Pin 4:DSR (receiving line for hardware handshake), only interface 2

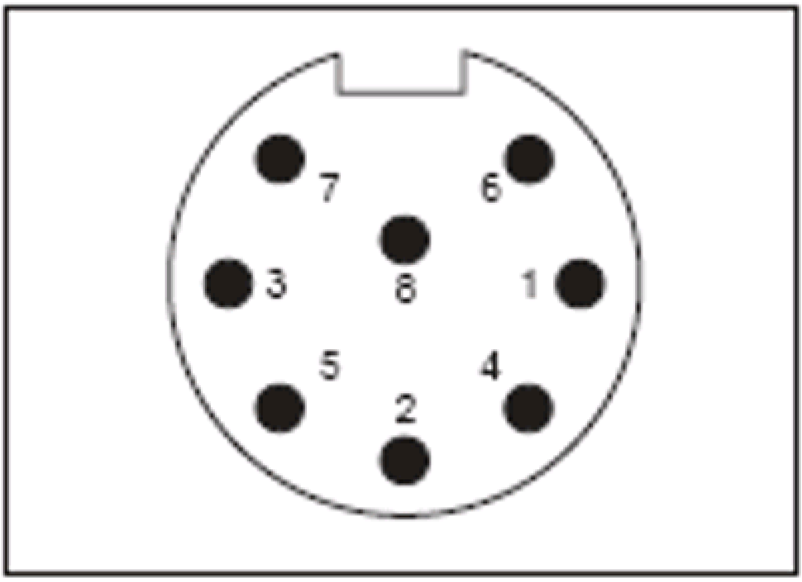
Pin 5:GND (signal ground)

Pin 6:DTR (transmission line for hardware handshake), only interface 2

Pin 8: V-ACCU (power supply from external storage battery, 6.3… 12VDC/200mA), only interface 1

Pin 9: Digital input for external contact (between Pins 5 and 9), only interface 1

Pin assignment RS232C Spider S (view on connector)



Pin 1: shielding

Pin 2: TxD (transmission line of the scale)

Pin 3:RxD (receiving line of the scale)

Pin 4:PONOFF (Power On/Off) for interface 1, DTR for interface 2

Pin 5: Digital input for external contact (between Pins 5 and 6), only interface 1

Pin 6:GND (signal ground)

Pin 7: V-ACCU (power supply from external storage battery, 6.3… 12VDC/200mA), only interface 1

Pin 8: BATLOW (external storage battery low) for interface 1, DSR for interface 2

Pin assignment RS232C Spider S (view on connector)

1 ———————— 1

2 ———————— 2

3 ———————— 3

4 ———————— 4

5 ———————— 5

6 ———————— 6

7 ———————— 7

8 ———————— 8

9 ———————— 9

Data cable 9-pole M/F, 1.8 m long, No. 00410024

This cable, for example, connects the Spider terminal with a PC or printer GA42. All pins are connected 1:1. This cable is equipped with a 9-pole Sub D plug (M) and a 9-pole Sub D socket (F).

Data cable 9-pole M/M, 1.8 m long, No. 21250066

This cable connects the Spider terminal with an optional second display. All pins are connected 1:1. This cable is equipped with two 9-pole Sub D plugs (M).

2.7 Scale Driver Nobel Elektronik E1TAD

2.7.1 General

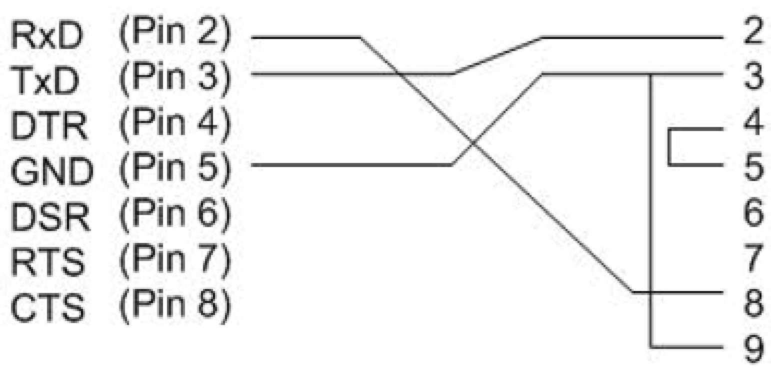
Description of the scale driver Nobel Elektronik E1TAD. Connection is established via a special RS232 cable. Scale values are requested by the terminal.

2.7.2 Interface

Using the RS232 connection, the host computer is directly connected with the scale via a serial cable.

Contact:	Short description	Meaning
2	RxD	Receive data
3	TxD	Transmit data
5	GND	Signal ground

Shop floor PC scale E-1-TAD



Default interface settings are:

9600 Baud

7 data bits

1 start bit

E parity bits

1 stop bit

There is **no** handshake. It must be guaranteed that the connected host computer is always able to hold a complete telegram in its receive buffer.

2.7.3 Settings nobel_e1tad.ini

[WAAGE_000]

COM=2,9600,7,E,1 → Settings for the serial interface

// The scale's communication address

Adr-Scale=02 → Communication address of the scale. If this address is not configured properly, the scale receives the request with a wrong address and recognizes that this request is not intended for it. Consequently, the scale sends the request to the next participant.

// Request parameters specific to scale.

Brutto_Scale=GV → gross value requested from scale

Netto_Scale=NV → net value requested from scale

V:EGR:GUTS=BRUTTO → Dialog ID including value requested from PCC.

Additional features

From version **7.2.1.1** on:

- The shop floor terminal can request a value (gross or net) via the PCC.

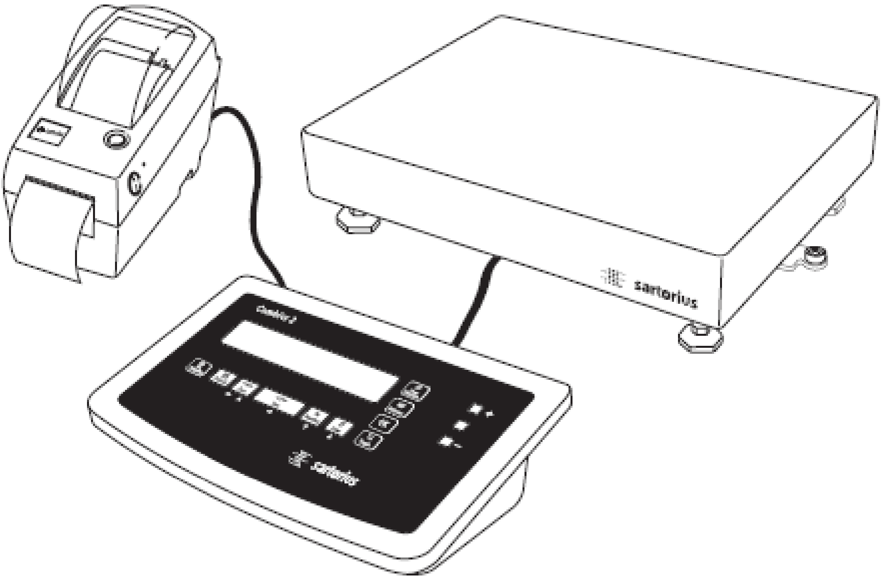
2.8 Scale Driver Sartorius Combics

2.8.1 General

The weighing unit consists of a display unit, scale pan and printers or computers that can be connected optionally. The driver supports the display units Sartorius Combics 1, Combics 1 plus and Combics 2 of the series CISL 1 / CISL 1 N / CISL 2 / CIS 1 / CIS 1 N / CIS 2 . The evaluation PC is connected to the display device via a serial RS-232 interface.

Required files:

- Sartorius_Combics.dll
- Sartorius_Combics.ini



2.8.2 Example for driver INI

[SERVICE]

info=sartorius_combics.dll

intervall=0

testmode=0

tracing=0

TraceLevel=5

ExecuteQueue=1

ThreadBaseId=100

[WAAGE_000]

COM=1,9600,8,O,1

; scale outputs acoustic signals

BEEP=ON

;BEEP=OFF

V:ERR=ERR

V:GEWICHT=NETTO

;V:GEWICHT=BRUTTO

V:EINHEIT=EINHEIT

POLL=0

POLL_I=500

2.8.3 Settings of Sartorius_Combics.ini

Serial interface

Interface parameters have been defined with 9600 Bd, 8 bits per character, one stop character and uneven parity. The free COM interface must be defined as the first parameter (COM=1,9600,8,O,1).

Acoustic signal

The scale may output an acoustic signal. If the parameter "BEEP" is set to "ON", a brief signal tone sounds, once weight has been transferred successfully to HYDRA. If an error occurred, the scale unit emits three short sounds. No signal will be sent if BEEP=OFF is set.

Weight parameter

A computer connected to the output unit can only request one weight value from the scale unit. This may be either a gross value or a net value depending on the value currently displayed. The driver module comparing the weight value with the Getval requested by PCC is informed about the value type. This driver module returns an error value in the ERR field if different weight types (gross and net) are used.

Example:

PCC asks for the gross weight by V:GEWICHT=BRUTTO. But the scale sends the net weight value. The driver returns the weight "V:Gewicht=0" and an error value with "V:ERR=BRU->NET". At the same time three short signal tones sound if BEEP=ON is set.

Possible error messages the driver may send are mentioned in one of the chapters that follow.

Weight unit

In the EINHEIT (UNIT) ID the scale driver sends the unit symbol of the scale. The following values are possible depending on the settings configured in the scale setup:

- EINHEIT=mg
- EINHEIT=g
- EINHEIT=kg

- EINHEIT=t or other units

All possibilities can be taken from a list in the manual about the scale.

Error messages

If the scale driver identifies an error in processing the PCC request, it returns the weight value "0" (e.g. BRUTTO=0) and classifies the error in the ERR field. The following error messages are possible:

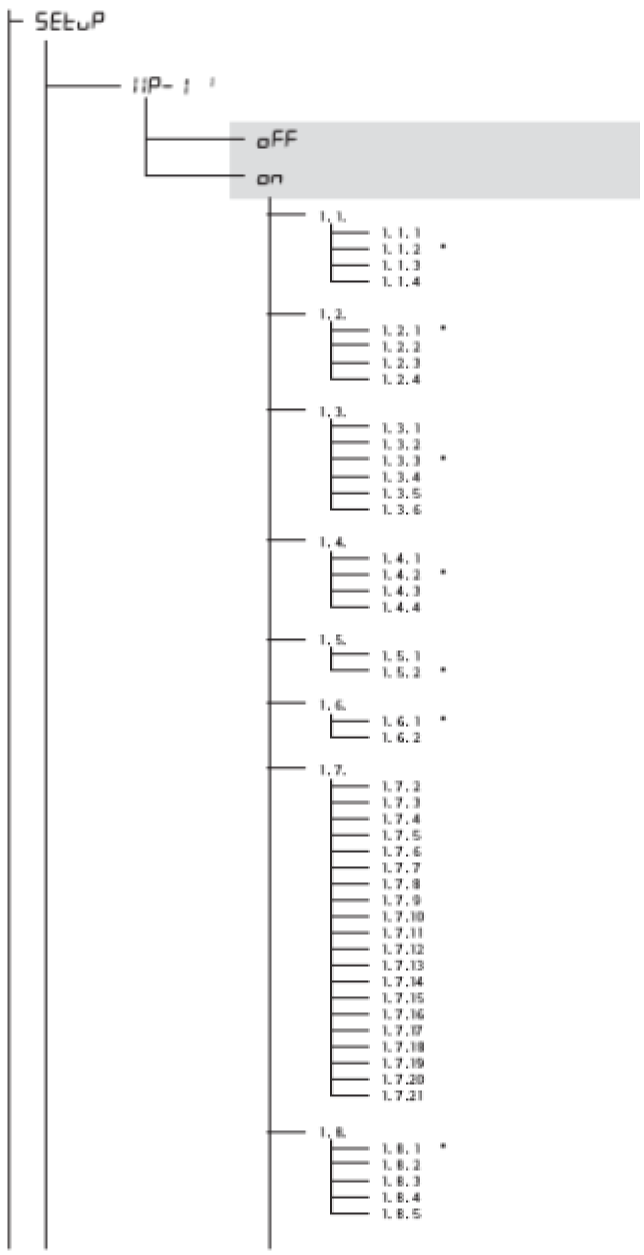
ERR=TIMEOUT	the scale does not answer
ERR=BRU->NET	PCC requests the gross value, but the scale sends the net value or vice versa
ERR=WAA-OFF	scale is on standby mode
ERR=ÜBERLAST	scale pan is overloaded
ERR=UNTER-L	load on scale pan insufficient
ERR=JUSTAGE	scale pan is being calibrated
ERR=NEGATIV	negative weight value displayed
ERR=54	if a numeric value is returned
	the corresponding error can be found in the manual dealing with the Sartorius scale.

If the weight value is still fluctuating during the weighing process, which is signaled by the missing unit symbol, the scale driver waits until the weight value is stable. If the value does not become steady within the TimeOut period, the driver sends: ERR=TIMEOUT

Scale setup

The following diagram shows the sections in the scale setup that must comply with the definitions in order to make sure the driver program works properly.

Setting the weight unit



Device parameters
Codes are requested if the code word is active.

Weighing platform 1
(display of this menu level)

OFF
ON

Adaptation to the installation site (adapt filters)
Very stable conditions
Stable conditions
Unstable conditions
Very unstable conditions

Filtering in use
weighing/balancing
dosing
Low degree of filtering
Without filtering

Stability range/fluctuation range
¼ increment (scale interval)
½ increment
1 increment
2 increments
4 increments
8 increments

Stability delay
Without delay
Short delay
Average delay
Long delay

Taring
Without stability
After stability

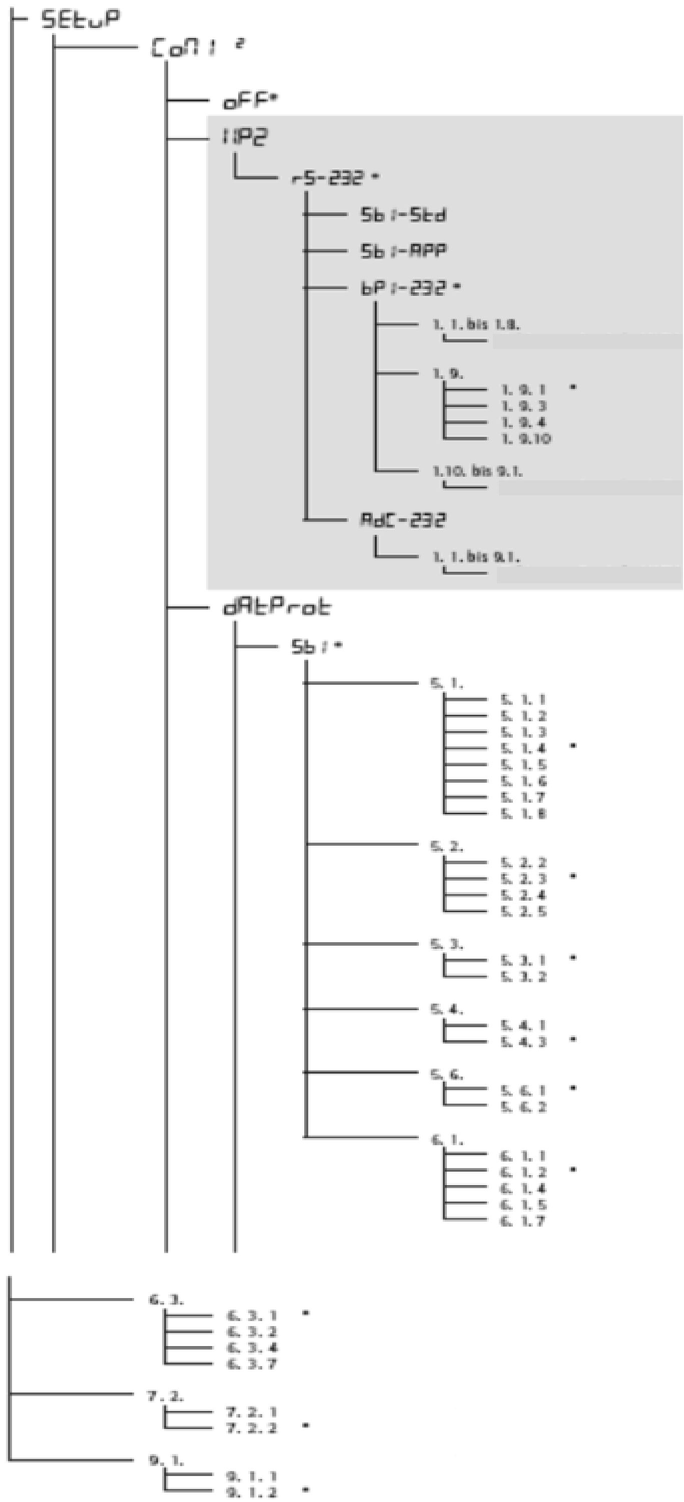
Autozero
On
Off

Weight unit 1
Gram / g
Kilogram / kg
Carat / ct
Pound / lb
Ounce / oz
Troy ounce / ozt
Tael Hong Kong / tlh
Tael Singapore / t
Tael Taiwan / tlt
Grain / GN
Pennyweight / dwt
Milligram / mg
Parts per Pound //lb
Tael China /tlc
Momme /mom
Carat /K
Tola /tol
Baht /bat
Mesgahl /MS
Ton /t

Display accuracy 1
All digits
Reduced by 1 digit when loads change
Reduce resolution by 1 scale interval (e.g. from 1g to 2g/5g to 10g)
Reduce resolution by 2 scale intervals (e.g. from 1g to 5g)
Reduced by 1 digit

Transfer parameters for the serial interface (arrows identify the parameters to be selected)

Arrows identify the parameters to be selected



Interface 1
(display of this menu level: 2)

OFF
Weighing platform 2
RS-232

SB1 standard version

SB1 calibration version

XBP1.232

Setup menu as with weighing platform 1

Calibration, adjustment
External calibration/adjustment; standard weight
External calibration/adjustment; weight can be selected (menu item 1.18.1)
Internal calibration/adjustment
Press key →T← to block scale

Setup menu as with weighing platform 1

ADC-232 ¹)

Setup menu as with weighing platform 1

Data logs

SB1 standard version

Baud rate

150	Baud
300	Baud
600	Baud
1200	Baud
2400	Baud
4800	Baud
9600	Baud
19200	Baud

Parity
Space (space character)
Odd
Even
None

Number of stop bits
1 stop bit
2 stop bits

Handshake operation mode
Software handshake
Hardware handshake, after CTS still 1 character

Number of data bits
7 data bits
8 data bits

Manual/automatic data output
Manual without stability
Manual after stability
Automatic without stability
Automatic with stability
Printing logs/protocols at computer (PC)

Time-dependent, automatic data output
1 display update/cycle
2 display updates
10 display updates
100 display updates

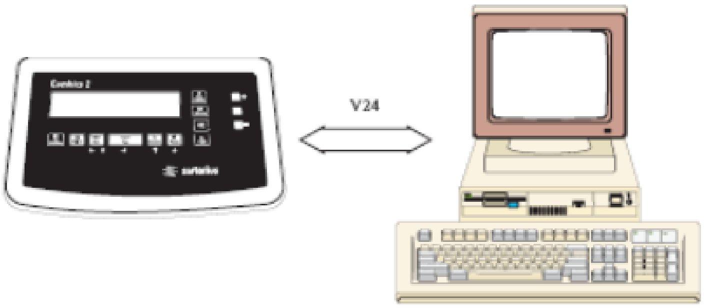
Data output: time format
For raw data: 16 characters
For other applications: 22 characters

Factory settings for the setup menu for COM1: SB1
Yes
No

¹) = menu depends on the connected scale/weighing platform

2.8.4 Connecting lines

Measurement system PC



Cable layout

Pin assignment for the cable from the measuring device to an RS-232-PC interface

25-pole D-Sub connector D-Sub socket

(Model CISLI1 / CISL2) 9-pole or 25-pole

Measurement device PC

Free cable end D-Sub socket

(Model CIS1 / CIS2) 9-pole or 25-pole

Measurement device PC